INCH-POUND
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SUPERSEDING
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PERFORMANCE SPECIFICATION

X-RAY APPARATUS SET, RADIOGRAPHIC, INDUSTRIAL, PORTABLE, 160KVCP

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains the technical and performance requirements for a light-weight, portable, industrial x-ray apparatus set. These x-ray apparatus sets are intended for use in a military aircraft maintenance environment, which may be at fixed military bases, deployed to austere forward operating locations, or naval vessels afloat. This specification provides for first article testing described herein.

1.2 CLASSIFICATION. The x-ray apparatus set should conform to the following types:

Type I - X-ray apparatus set with a panoramic, liquid-cooled tube head.
Type II - X-ray apparatus set with a directional, liquid-cooled tube head.
Type III - X-ray apparatus set with a, directional air-cooled tube head.
Type IV - X-ray apparatus set with a panoramic, air-cooled tube head.
Type V - X-ray apparatus set with a small diameter, directional, liquid-cooled tube head.

1.2.1 Part Identification Number (PIN). The following part or identifying number (PIN) procedure is for government purposes and does not constitute a requirement for the contractor. Example of part or identifying number:

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- Type (See 2): 1 – Type I; 2 – Type II; 3 – Type III; 4 – Type IV; 5 – Type V
- Specification Number
- Prefix

Comments, suggestions, or questions on this document should be addressed to: AFLCMC/EZSS, 2145 Monahan Way, Area B, Bldg 28, Wright-Patterson AFB, OH 45433 or email to ENGINEERING.STANDARDS@US.AF.MIL. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.2.2 National Stock Number (NSN). Interchangeable & Substitutable Master NSN Cross Reference:

	NSN	CAGE	Part Number
a.	6635-01-430-2575	RN81348	M32133-1
b.	6635-01-394-5926	RN81348	M32133-2
c.	6635-01-442-6915	RN81348	M32133-3

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-28800 - Test Equipment for use with Electrical and Electronic Equipment, General Specification For

(Copies of these documents are available online at https://assist.dla.mil/.)

2.2.2 Other government documents, drawings, and publications. The following government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those specified in the solicitation or contract.

ENVIORNMENTAL PROTECTION AGENCY REGULATIONS

Environmental Protection Agency list of pollutants/toxins.

(Copies of these documents are available online https://www.epa.gov/.)

CODE OF FEDERAL REGULATIONS

21 CFR 1040.10 - Laser Products

(Copies of this document is available online at: https://www.ecfr.gov/.)

FEDERAL ACQUISITION REGULATIONS

FAR 23.403 - Federal Acquisition Regulations (FAR)

(Federal Acquisition Regulations may be obtained at: https://www.acquisition.gov/.)

FEDERAL STANDARDS

FED-STD-376 - Federal Standards

(Copies of this document is available online at https://www.nist.gov/.)

Non-Government publications. The following documents form a part of this document to the

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extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E1165-12	-	Standard Test Method for Measurement of Focal Spots of Industrial X-ray Tubes by Pinhole Imaging
ASTM-E1742/ E1742M	-	Standard Practice for Radiographic Examination
ASTM-E2445/ E2445M	-	Standard Practice for Performance Evaluation and Long-Term Stability of Computed Radiography Systems

(Copies may be obtained at: www.astm.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 Article 500	-	National Electric Code

(Copies of this document may be obtained at: www.nfpa.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the solicitation (see 6.2), two samples shall be subjected to a first article inspection in accordance with 4.

3.2 Description. The x-ray apparatus set shall consist of an x-ray tube head and control unit. Types I, II and V, shall include a liquid cooling unit. The x-ray apparatus set (all types) shall be supplied with the following accessories: Transit cases, cable assemblies, lightweight laser pointer, interlock connector, safety lock keys, tube head stand, and manuals. Types I, II and V shall include coolant hoses, as required.

3.3 Foreign object damage. The x-ray apparatus set shall be designed to prevent loss of components, which could generate foreign object damage to weapon systems or engines. Connector protective covers shall have a high strength cable attached to the cover and component.

3.4 Interchangeability. The apparatus set major components shall be interchangeable with other apparatus set major components of the same make and model series with no degradation of performance. The interchangeability of major components shall not change the output of the x-ray source by more than ±3 percent.

3.5 Connectors. The apparatus set major components shall have bayonet or quick turn type connectors. Each connector shall be color coded and unique so cables fit the intended connector. Mating connectors shall maintain color coding.

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3.6 Strain relief. Electronic cables shall incorporate strain relief features.

3.7 Connectivity. Wireless capabilities shall be disabled.

3.8 X-ray output. The X-ray output shall be adjustable in 5 kilovolt constant potential (kVcp) or less steps minimum over the range of 20 to 160 kVcp, and in 0.1 milliampere (mA) steps over the range of 1 to 5.0 mA, and there shall be no more than 2 percent ripple. Maximum power at the anode shall be no less than 750 watts.

3.8.1 High voltages. All high voltages shall be generated in the tube head.

3.8.2 Duty Cycle. The duty cycle for liquid-cooled sets shall be continuous at maximum output from 0 to 120 degrees Fahrenheit, and for air-cooled sets, 100 percent at maximum output for one hour at 85 degrees Fahrenheit.

3.8.3 Operating Temperature. The X-ray apparatus set shall be capable of operating in temperatures from 0 to 120 degrees Fahrenheit.

3.8.4 Storage temperature. The x-ray unit shall be capable of storage temperatures from 0 to 150 degrees Fahrenheit**.**

3.8.5 Power Requirements.

3.8.5.1 Input power. Input power shall automatically adapt from 100-130 Volts Alternating Current (VAC), 50/60 Hertz, 20 Ampere maximum current, to 200-250 VAC, 50/60 Hertz, 10 Ampere maximum current, and include automatic adaptability to portable generator power.

3.8.5.2 Overvoltage/overcurrent protection. The components shall be equipped with circuit breakers to prevent damage to components in the event the voltage tolerances and/or proper current levels are exceeded. The circuit breakers shall provide protection to these components when large voltage or current surges are experienced.

3.8.5.3 Transient-State Conditions. The apparatus set shall not experience degradation of performance after the conditions defined in MIL-PRF-28800, section 3.

3.8.5.4 Voltage Frequency protection. The apparatus set shall be protected from input voltages and frequencies that exceed the steady state envelope conditions in 3.8.5.1. If such events should occur the apparatus set shall protect itself by automatically disconnecting itself from the power source until the discrepancy is eliminated.

3.8.6 Tube head.

a. Features the x-ray tube head shall be end grounded and gas insulated, with a constant potential system. Types II, III and V shall include a side port beryllium window. The tube head shall have a pressure gage, pressure relief valve, and low pressure cut-out switch which will prevent operation below a specified pressure.

b. For directional tube heads, the effective focal spot shall be no greater than 0.063 inch (1.6 mm), FS5 class or better, in accordance with ASTM-E1165-12.

c. X-ray tube window material and thickness.

1. Panoramic Tube heads: 0.02 inch thick or less of nickel or equivalent filtration.

2. Directional Tube heads: 0.031 inches thick or less of beryllium or equivalent filtration.

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d. A thermal temperature switch shall be provided to prevent operation if the tube head overheats beyond stated design temperatures.

e. The panoramic tube head shall have a 360 degree cone of radiation output (field of view).

f. The directional tube head shall have a maximum 40 x 60 degree cone of radiation output (field of view).

3.8.7 Volatile ignition. Type I, II, and V x-ray tube heads shall not be a source of ignition of volatile atmospheres as defined by the National Fire Protection Association (NFPA) 70 Article 500 for Class 1, Division 2, Group D during both operation and non-operation of the tube head.

3.8.8 X-ray exit port. The type V tube head shall have the x-ray exit port near the end of the tube head with a diameter not to exceed 3 inches and extend not less than 3 inches from the tube head if applicable.

3.9 Control unit.

a. The control unit shall incorporate a safety key switch and supplied with two keys. X-ray production shall cease if the key is removed while x-rays are produced.

b. The control unit shall incorporate an emergency stop button that when activated instantly ceases x-ray generation. X-rays can only be generated after the button is reset and the normal x-ray generation sequence is followed.

c. The control unit timer shall be adjustable from 0.1 to 60 minutes in 0.1 minute or smaller increments.

d. The control unit shall have microprocessor-based diagnostics and display apparatus fault messages and interlock safety failure messages. X-ray production shall cease if fault or safety condition occur.

e. Faults and diagnostic monitoring shall include as a minimum, open interlock, emergency stop, tube head temperature and pressure, mains voltage, arc fault, kV and mA fault, short circuits, and tube cable disconnect. Type I, II and V shall include coolant flow fault.

f. The control unit shall include a continuous display to indicate as a minimum, tube head voltage set, tube head voltage applied, tube head current set, tube head current applied, time set, and time elapsed/remaining.

g. The control unit shall be equipped with circuit breakers (see 3.8.5.2).

h. The control unit shall display settings and parameters in inch-pound units, as applicable, and markings in the English language.

i. The control unit shall incorporate automatic warm up cycle(s) based on time of last use and parameters of exposure.

j. Two, 50 foot cables, with strain relief, capable of connecting end-to-end to form 100 feet of cable shall be furnished to connect the control unit to the tube head.

k. Two mains power cables shall be furnished, 1, 10 foot and 1, 50 foot.

l. The x-ray apparatus set must be compatible with X-ray safety interlock system: NSN 6635-01-508-1519RN and supplied with a 10 foot interface cable.

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3.10 Cooling unit. The cooling unit for liquid-cooled types shall be a self-contained, closed circuit (of a closed-loop design) and shall not contain any toxic chemicals, hazardous substances, or ozone depleting compounds as listed in the Environmental Protection Agency list of pollutants/toxins.

3.10.1 Self-sealing connectors. Cooling hoses and unit shall be supplied with self-sealing connectors. Supply/return hoses shall be clearly color coded or connectors provided for unique fit for each connection. Two, 50-foot cooling hose sets shall be supplied.

3.10.2 Cooling unit. The liquid coolant unit shall be capable of maintaining the x-ray tube head within normal operating temperatures when the x-ray tube head is 100 feet in elevation above the cooling unit.

3.10.3 Cooling unit flow sensor. The cooling unit shall incorporate a coolant flow sensor to prevent overheating of the tube head.

3.11 Physical dimensions and weight. The equipment must not exceed the following dimensions or weights.

3.11.1 Tube head. The tube head shall have a maximum diameter of 8-1/2 inches and maximum weight of 40 pounds without hoses or cables. For tube heads with detachable lifting handles/rings, the handles/rings shall be removable using common tools without other component disassembly.

3.11.2 Control unit. Maximum weight 35 pounds when not enclosed in the ruggedized transit cases, with no dimension exceeding 18 inches.

3.11.3 Cooling unit. Maximum weight 55 pounds when not enclosed in the ruggedized transit cases, with no dimension exceeding 18 inches.

3.12 Accessories.

3.12.1 Transit case. Ruggedized transit case(s), suitable for cargo aircraft transport, shall be provided for all major subassemblies and accessories, and shall be made of corrosion-resistant material. No case with contents shall exceed 70 pounds.

3.12.1.1 Transit cases design. The ruggedized transit cases shall be watertight (just over the top of case only) and be designed to provide protection against shock, vibration, and environmental conditions per MIL-PRF-28800, Class 2, including vibration test, mechanical shock test, transit drop test, and water resistance test.

3.12.2 Soft case. A “soft” case, made of foam- lined heavy-duty nylon fabric, shall be provided to protect the tube head when not in use or in the ruggedized shipping containers.

3.12.3 X-ray shield. An x-ray shield shall be supplied that fits over the x-ray window which reduces x-ray output, to less than 2 milliroentgens per hour at 39.37 inches (1 m) when at maximum output.

3.12.4 Laser pointer. A lightweight laser pointer, integral to or attachable to the tube head, useable to 50 feet, which complies with 21 CFR 1040, Class III a, maximum shall be provided to align with the beam exit point. The laser pointer shall be easily affixed and removed without tools if not integral to the tube head.

3.12.5 Handles. The transit cases, control unit, and cooling unit shall have collapsible handles with permanent stops to prevent collapsing while handles are in use.

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3.12.6 Operation and maintenance instruction. One hard copy and one electronic (CD-ROM /DVD) copy of the commercial operation and maintenance instructions, including error and diagnostic codes or messages, and calibration procedures and adjustments, as applicable, shall be shipped with each item. The documentation shall be releasable for third-party repairs. Documentation shall be in the English language. Two paper and two electronic (CD-ROM /DVD) copies shall be sent to the USAF Equipment Specialist and Cognizant Engineer.

3.12.7 Parts kit. A parts kit shall be included to contain adapters, and expendable parts that are expected to be consumed in one year needed to effect user-serviceable repairs.

3.12.8 Tube head stand. The tube head stand shall conform to the following:

a. The tube head stand shall be designed to hold and position any of the tube heads without damage to the tube head.

b. The tube head stand shall incorporate locking mechanisms for each adjustment device.

c. The tube head stand shall consist of a tripod, adjustable tube head mount and tube head cradle assembly. Each leg shall be separately adjustable. The tube head shall possess an elevator mechanism which is operated by a hand wheel that raises and lowers the tube head.

d. The tube head cradle assembly shall secure the tube head to the head mount with clamping devices.

e. The tube head elevator mechanism shall be able to elevate the tube head at least 72 inches from the ground, when the tube head is in the horizontal position.

f. The head mount shall permit 360 degree rotation in the horizontal plane and tilting of the tube head from 45 degree below horizontal to 90 degree above horizontal (absolute vertical) in the vertical plane.

g. The total weight of the tube head stand shall not exceed 40 pounds.

3.12.9 Accessory items. Additional accessory items. A listing of available accessory items, extended warranties, and standard repairs shall be provided to the Contracting Officer.

3.13 Operation manual. One paper copy and one electronic copy of the commercial operation and maintenance instructions, including calibrations procedures and adjustments, as applicable, shall be shipped with each apparatus. All documentation shall be releasable for third-party repairs. A draft copy shall be provided with the production evaluation article for engineering approval. Two paper and two electronic copies of the final version shall be sent to the USAF Equipment Specialist and Cognizant Engineer.

3.14 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

3.15 Toxic chemicals, hazardous substances, and ozone-depleting chemicals. The use of toxic chemicals, hazardous substances, or ozone-depleting chemicals shall be avoided, if feasible.

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3.16 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds unit, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

3.17 Warranty. A minimum one-year warranty, to include parts, labor, and shipping, shall be included against manufacturing defects.

4. VERIFICATION

4.1 First article tests. Two x-ray apparatus sets shall pass any of the requirements and conformance inspections described in this document, regardless of the bidder's past performance. The inspection will not exceed 60 days. One system shall be tested to the requirements of 4.5, Conformance Inspection, by either a U.S. Government or independent commercial laboratory. The other system shall be operational tested at a Department of Defense Aircraft Maintenance Facility in the continental United States. In the event minor deficiencies/problems are found, any corrective action will be negotiated/approved by the Contracting Officer. One additional on-site visit by the bidder to repair/adjust the equipment, plus up to two shipments of user installable replacement parts, shall be permitted during the First article test for each of the two x-ray apparatus sets. The x-ray apparatus shall be returned to an operational condition within seven days after notification to the bidder.

4.2 Product conformance. The products provided shall meet the salient characteristics of this specification, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The government reserves the right to require proof of such conformance.

4.3 Market acceptability. The following market acceptability criteria are necessary to document the quality of the x-ray apparatus to be provided under this document.

4.3.1 Market longevity. The manufacturer of the x-ray apparatus must have been producing x-ray apparatus sets for at least four years.

4.3.2 Number of units sold. The company must have sold at least 100 x-ray apparatus in the commercial market place over the past three years with no major defects within the last two years. A major defect is a defect, which affects the performance of the apparatus and requires an engineering change to the apparatus' original configuration in order for the apparatus to perform to the requirements of this document.

4.3.3 Location of repairs. Repairs shall be provided in the contiguous U.S. and shall have an average repair time of two calendar weeks or less.

4.4 Pre-award surveys. The government reserves the right to conduct pre-award surveys at the bidder's facilities. Government technical representatives will advise the Contracting Officer of the bidder's capability to produce x-ray apparatus sets in accordance with this specification to an acceptable quality level.

4.5 Conformance inspection. The x-ray apparatus shall pass any of the requirements of this paragraph, conformance inspection, and paragraph 3, requirements, by either a U.S. Government representative, independent commercial laboratory and/or an operational test at a Department of Defense Aircraft Maintenance Facility in the continental United States. In the event minor deficiencies/problems are found, any corrective action will be negotiated/approved

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by the Contracting Officer. One additional on-site visit by the bidder to repair/adjust the equipment, plus up to two shipments of user installable replacement parts, shall be permitted during the conformance inspection test for each of the x-ray apparatus set(s). The x-ray apparatus shall be returned to an operational condition within seven days after notification to the bidder.

4.5.1 Examination of Components. Visually inspect to insure the x-ray apparatus contains the following components and meets the requirements as specified in 3:

a. X-ray Tube head, Type per contract order.

b. Control Unit, digital microprocessor-based.

c. Liquid Cooling Unit, applicable to Types I, II and V.

d. Operation & maintenance manual.

e. Ruggedized transit case for tube head, applicable for proper Type.

f. Ruggedized transit case for control unit.

g. Ruggedized transit case for cooling unit applicable to Types I, II and V.

h. Ruggedized transit case for cables, hoses, and accessories.

i. 2 keys for control unit safety lock.

j. Control cable assembly – 2, 50 foot forming 100 feet with strain relief.

k. Cooling unit cable assembly – 2, 50 foot forming 100 feet with strain relief (Types I, II and V) each with self-sealing termination.

l. Safety Interlock cable assembly – 10 feet with strain relief.

m. Lightweight laser pointer (removable), useable to 50 feet.

n. Tube head stand.

4.5.2 Dimensional Inspection. Measure equipment dimensions and weight to ensure all components meet the requirements specified in 3.

4.5.3 Operational Tests.

4.5.3.1 Timer test. Test control unit timer to ensure 0.1 to 60 minutes in 0.1 minute steps.

4.5.3.2 Unit display. Ensure control unit continuously displays tube head voltage – actual and set; tube head current – actual and set; exposure time – elapsed/remaining and set.

4.5.3.3 Control unit test. Test control unit to ensure continuously adjustable over the range of 20-160kVcp in 5kV steps minimum and 1-5.0 mA in 0.1mA steps. Record mA at 20kV and 160kV. Record kV at 1mA and 5mA.

4.5.3.4 Diagnostic test. Conduct tests, as applicable, to check diagnostic monitoring.

4.5.3.5 Focal spot test. Using ASTM-E1165-12, pinhole camera method, measure the effective total focal spot size for directional tube heads, to ensure it is no greater than 0.063 inch (1.6mm).

4.5.4 Performance. The following tests are performed using an USAF approved computed radiography (CR) system using an ultra-high resolution, approved computed

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radiography imaging plate scanned at 50 micron resolution.

4.5.4.1 X-ray capability test. The x-ray unit shall be capable of showing a 0.010 inch hole in a 0.010 inch thick aluminum IQI (per ASTM-E1742/1742M) placed on a 1 inch thick test block of 1000 series aluminum at a source to detector distance of 36 inches. The IQI and imaging plate shall be perpendicular to the x-ray beam.

4.5.4.2 X-ray noise ratio. The x-ray unit shall be capable of producing an image showing an EPS value of 1.28 percent or less, or signal to noise ratio of 150 or better, reference ASTM-E2445/E2445M.

4.5.4.3 X-ray image. The x-ray unit shall be capable of producing an image showing 8 line pairs per millimeter and three steps of the contrast sensitivity gauge, reference ASTM-E2445/E2445M.

4.5.4.4 Source to film distance. The x-ray unit shall be capable of producing an image showing a linear pixel value ratio greater than 2.7 between aluminum foil of 0.003 inches and 0.03 inches thickness at 36 inches source to film distance.

4.5.4.5 X-ray duty cycle. (For Type I, II and V) Ensure the x-ray apparatus set is capable of operating with the tube head 100 feet above the cooling unit. Operate the unit at maximum output for 10 successive operating cycles of 5 minutes on and 1 minute off. Any overheat condition is evidence of failure.

4.5.4.6 Cooler flow sensor. (For Type I, II and V) Ensure the cooler unit flow sensor is operational by operating the apparatus set at 100kV/5mA and disconnecting the cooler hose. If the apparatus set does not cease x-ray generation in 10 seconds and/or is capable of further operation until the coolant flow is restored, this is evidence of failure.

4.5.4.7 X-ray safety interlock. Ensure the x-ray apparatus set is compatible with x-ray safety interlock, NSN 6635-01- 508-1519RN.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These x-ray apparatus sets are intended for field and depot use to perform radiographic inspections of metallic and nonmetallic, aerospace and non-aerospace equipment.

6.2 Acquisition requirements:

a. Title, number, and date of this specification.

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b. Part or Identifying Number (see 1.2.1).

c. First article inspection requirements (see 3.1 and 4.1).

d. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Constant potential
Deployable
Penetrameter
Tube head

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – AS
Air Force – 171

Preparing activity:
Air Force – 171
Agent:
Air Force – 110

Review activities:
Navy – CG
Air Force – 71, 84
DLA – GS

(Project 6635-2021-002)

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